

Harris Ahmad

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EDUCATION

University at Buffalo, PhD Computer Science

January 2025 - May 2028

- **Teaching Assistant:** Modern Networking Concepts
- Available for **Summer 2027** internships and returning to the program afterwards, with completion expected **May 2028**.

YC AI Startup School: [\[Link\]](#)

July 2026

- Selected from a pool of **30,000** applications to attend **YC AI Startup School** in the **SF Bay Area**.

Lahore University of Management Sciences, BS Computer Science

September 2020 - May 2024

- **Teaching Assistant:** Object Oriented Design in C++, Distributed Systems

PROFESSIONAL EXPERIENCE

Research Software Engineer, University at Buffalo (SUNY)

January 2025 – Present

- Built a distributed transaction **reordering and coordination** layer over **Apple's FoundationDB** and **CockroachDB** in **Go**, reducing aborts, lock contention, and tail latency by **70%**; **in preparation for USENIX OSDI 2027**.
- Prototyped **primary-backup**, **chain replication**, and **distributed two-phase locking** in **Go** to stress-test protocol behavior for **collaborative AI** workloads.
- Building a **coordination-free concurrency protocol** for **microservices** in **Java**: keeps distributed transactions correct without the central coordinator and global row locks **Apache Seata** holds across a commit, which serialize conflicting transactions and add a round trip to every one; **in preparation for USENIX OSDI 2027**.

Software Engineer, Linq.io

July 2024 - December 2024

- Engineered a system-level **playground and diagnostics platform** for an **asset management system**, supporting **workflow simulation**, **feature validation**, **API testing**, and backend **observability** through **structured logging**, **distributed tracing**, **fault isolation**, and **real-time system monitoring**.
- Shipped an on-demand **media-generation API (FastAPI, MongoDB)** that cut page load by **60%** by generating **asset thumbnails** as users browsed **10K+** inventories.
- Integrated **OAuth 2.0 SSO** for **enterprise** accounts so customers could sign in to the asset-management product with **existing corporate credentials**.
- On call **twice a week** for **production pipelines**, resolving client issues tracked in **Jira**; took part in **weekly code reviews** with senior engineers.

System Engineer, Its IT Group

May 2024 - December 2024

- Redesigned the backend (**Node.js**, **MongoDB**) of an **AR** app for **VTuber/karaoke 3D avatars**, using **WebSockets** for real-time comments, notifications and likes across **5K+** daily active users.
- Implemented a **3D model pipeline** processing **LiDAR** scans, building a **USDZ to GLB** converter for **Unity** integration that cut manual measurement time by **80%**.

Software Engineer, Interactive Media Lab LUMS

October 2023 - March 2024

- Built **Awaaz-e-Sehat**, a **HIPAA-aligned e-health platform** on **Flask** and **AWS Lambda**, improving API latency **5s** → **500ms** for **20K+** daily active users.
- Productionized **Whisper + GPT-4** transcription into structured clinical notes (**95%** accuracy, **1K+** recordings/day); **Redis** caching reduced API latency by **~30%**.

Research Software Engineer, Network and Systems Group LUMS

May 2022 - December 2023

- Released a **public dataset** and toolchain covering **17.6K** videos and **46.6K** ads across **8** countries with **per-video buffer telemetry** — the **first large-scale YouTube corpus** with that coverage; published at **[ACM The Web Conference 2024 \(WWW '24\)](#)**.
- Worked in **information retrieval** at scale: built the collection and analysis toolchain behind the corpus and released it publicly alongside the dataset.
- Measured how **YouTube** main-video **bitrate** drives **buffer loss** across resolutions (**720p** ~**3x** higher **latent buffer loss** than **360p**; **10.1 MB** vs **3.4 MB**).

PROJECTS

BlastRadius: [\[Link\]](#) [\[PyPI — 336 installs\]](#)

August 2026 - Present

- **Cross-repo dependency memory for coding agents** — tells an agent which *other* repositories break before it bumps a shared **Docker** image, **Terraform** module, **GitHub Action**, **Helm** chart, or **npm** package. Ships as an **MCP**

server (blastradius-mcp) on PyPI and the Claude Code plugin marketplace.

- Fires **unprompted** from **PreToolUse** hooks on **Bash** as well as **Read/Edit**, catching `npm install` rewrites no editor sees; an **OSV** daemon matches advisories by **semver range** and **CVSS**, surfacing shared **CVEs** with **no fixed version** that an exact pin cannot escape.
- Ships **no manifest parsers**, delegating extraction to the model and measuring it: a planted **6-repo** corpus scores **39/39** recall at **39/39** version specs intact and **0** false positives; **376** tests against **3.6K** lines across **16** modules.
- Serves the index over the **Model Context Protocol** — three **MCP** tools on the official SDK over **stdio** — while the hooks push the same facts unasked, because a tool only fires if the model elects to call it and a hook fires regardless.
- Scores advisories from **OSV.dev** by computing **CVSS v3.1** base scores from the raw vector, and decides applicability with three-valued **semver** and **PEP 440** range evaluation that counts *unknown* as affected rather than clean.
- Indexes six artifact types in **SQLite** tuned for concurrent readers — **WAL**, thread-local connections for a pooled server, additive migrations, and ranking pushed into **SQL** with cached aggregates maintained by a **CTE** update.
- Measures a model's extraction instead of trusting it: an adversarial corpus of **multi-stage Docker** aliases, **ARG**-templated images and a **FROM** inside a heredoc, graded on recall, false positives and version-spec fidelity, failing on any of the three.

deadpush: [Link] [PyPI — 502 installs]

June 2026 - Present

- **Guardian daemon that stops an AI coding agent destroying your work** — checkpoints every file a destructive command would touch, then blocks the command. A **content-addressed SHA-256** journal with two-phase preflight restore; a **48-run** ablation doubled block rate **0.33 to 0.67** and work-preserved **0.92 to 1.00** at zero false positives.
- Red-teamed and closed same-UID **git hook** bypasses — forged **refs/remotes**, **core.hooksPath** hijack, out-of-band **commit-tree** — with an **empty-tree** scan and daemon **HEAD** polling; **50** tests, **11** techniques.
- Exposes **22 MCP tools** over hand-written **JSON-RPC 2.0** on **stdio**, and ships a transparent **MCP proxy** that intercepts `tools/call` before a downstream server runs, so the guardrails apply to tools it does not own.
- Enforces on **Linux** through **fanotify** bound with **ctypes**, and isolates a session with **bubblewrap** namespaces or a macOS **Seatbelt** profile behind one backend interface.
- Keeps a **tamper-evident hash-chained audit log** anchored on a genesis hash with its own `verify` command, exports findings as **SARIF 2.1.0** for GitHub's security tab, and reaches hidden payloads by **import-graph** traversal.
- Holds **13** formally specified guarantees, each naming the test that proves it and the attack that defeats it, over **679** tests in **113** classes.

SKILLS

- Languages: Python, Go, Java, JavaScript/TypeScript, C, C++, Haskell, PowerShell, Bash
- Systems and data: distributed systems, query processing and optimization, Linux/Unix, FoundationDB, CockroachDB, PostgreSQL, MongoDB, Redis, Elasticsearch
- Backend and APIs: FastAPI, Flask, Django, Node.js (Express), gRPC, GraphQL, WebSockets, OAuth 2.0 / SSO
- Cloud, ML: AWS (Lambda, DynamoDB, S3), Docker, K8s, GCP, TensorFlow, scikit-learn, LangChain, NLP
- Frontend and tooling: React.js, Vue.js, Next.js, D3.js, Streamlit, HTML/CSS, GitLab CI, Jenkins, Ansible, Jira